

Maro HW 2



$$\dot{k} = s \cdot (w + rt) - \delta k$$

$$Y_i = (A L_i)^{1-\alpha} K_i^\alpha \quad A = a \left(\frac{1}{N} \sum_{i=1}^N K_i \right)^b$$

1. In factor market, total quantity demanded by firm = total quantity supplied by all consumers.

$$L = \sum_i^N L_i \quad K = k \cdot L = \sum_{i=1}^N K_i$$

2. Derive r and w . As model parameters and average capital labor ratio

$$w = \frac{\partial (A L_i)^{1-\alpha} K_i^\alpha}{\partial L_i} = (1-\alpha) A^{1-\alpha} L_i^{-\alpha} K_i^\alpha =$$

$$\begin{aligned} (1-\alpha) \left[a \left(\frac{K}{N} \right)^b \right]^{1-\alpha} \left(\frac{K_i}{L_i} \right)^\alpha &= (1-\alpha) a^{1-\alpha} \left(\frac{K}{N} \right)^{(1-\alpha)b} K_i^\alpha \\ &= (1-\alpha) a^{1-\alpha} \left(\frac{L \cdot k}{N} \right)^{(1-\alpha)b} K_i^\alpha \\ &= (1-\alpha) a^{1-\alpha} \left(\frac{L}{N} \right)^{(1-\alpha)b} k^{(1-\alpha)b + \alpha} \end{aligned}$$

$$Y_i = (A L_i)^{1-\alpha} K_i^\alpha \quad L \cdot k$$

$$\begin{aligned} \frac{\partial Y_i}{\partial K_i} &= \alpha A^{1-\alpha} L_i^{1-\alpha} K_i^{\alpha-1} = \alpha \left\{ a \left[\left(\frac{L \cdot k}{N} \right)^b \right]^{1-\alpha} \right\} K_i^{\alpha-1} \\ &= \alpha a^{1-\alpha} \left(\frac{L}{N} \right)^{b(1-\alpha)} k^{b(1-\alpha) + (\alpha-1)} = \alpha a^{1-\alpha} \left(\frac{L}{N} \right)^{b(1-\alpha)} k^{(1-\alpha)(b-1)} \end{aligned}$$

ECON*6020 Macroeconomic Theory I
Assignment 2 – Fall 2026

Instructor: Stephen Kosempel
Due Date: Friday October 9, 2026

Consider an economy that consists of L identical consumers and N firms. Each consumer supplies 1 unit of labour to the firms each period. The wage rate per worker is given by w . Consumers also rent capital to firms. The rental rate of capital is given by r . The budget constraint for the representative consumer is given by

$$\dot{k} = s \times (w + rk) - \delta k,$$

where k denotes the capital-labour ratio (K/L), s is the savings rate, and δ is the depreciation rate. There is no population growth.

Each producer i creates output via a Cobb-Douglas constant returns to scale production function given by

$$Y_i = (AL_i)^{1-\alpha} K_i^\alpha,$$

where A denotes the state of technology. Arrow (1962) has endogenized the growth process in the Solow-Swan model by assuming that a producer that increases its physical capital learns simultaneously how to produce more efficiently. The positive effect of experience on productivity is called learning-by-doing, or in this case learning-by-investing. We will attempt to capture Arrow's concept of learning-by-investing by assuming that the state of technology is a function of the average capital stock in the firms,

$$A = a \left(\frac{1}{N} \sum_{i=1}^N K_i \right)^b, \text{ where } a > 0 \text{ and } b \geq 0.$$

Firms in choosing their own K_i and L_i treat A as given. Therefore, there is an externality to capital accumulation.

- (a) State the market clearing conditions in the factor markets.
- (b) Derive the competitive equilibrium real wage rate and real rental rate of capital as a function of only the model's parameters and the average capital-labour ratio in the economy.
- (c) Show that for $\underbrace{b > 0}_{\text{learning-by-investing}}$ the economy as a whole has increasing returns to scale in capital and labour.
- (d) Describe the transition dynamic to the steady-state for: (i) $b < 1$, (ii) $b > 1$, and (iii) $b = 1$.

(e) For $b=1$ what is the steady-state growth rate of the economy?

(f) Set $\alpha=1/3$, $a=1$, $\delta=10\%$, $L/N=1$, and $s=25\%$. Derive the steady-state level of capital for $b=0$. Starting with an initial capital-labour ratio that is $1/4$ the steady-state level, simulate a series using MATLAB for the economy's growth rate of income per capita for the next 100 years. Now set $b=1/2$ and redo the experiment, but do not change the starting value. Identify the key differences in the two growth paths. Which version do you think fits the data better: $b=0$ or $b=1/2$?

(g) For the $b=0$ case solve for the Golden Rule level of capital. Is $s=25\%$ dynamically efficient?

c)

$$\begin{aligned}
 Y_i &= (A L_i)^{1-\alpha} K_i^\alpha \\
 Y &= \sum_{i=1}^N (A L_i)^{1-\alpha} K_i^\alpha \\
 &= \sum_{i=1}^N A^{1-\alpha} L_i \cdot L_i^{-\alpha} K_i^\alpha = \sum_{i=1}^N A^{1-\alpha} L_i \left(\frac{K_i}{L_i} \right)^\alpha \\
 &= \sum_{i=1}^N A^{1-\alpha} K_i^\alpha L_i \\
 &= A^{1-\alpha} K^\alpha \sum_{i=1}^N L_i \\
 &= A^{1-\alpha} K^\alpha L
 \end{aligned}$$

$$\begin{aligned}
 &= A^{1-\alpha} \left(\frac{\sum_{i=1}^N K_i}{N} \right)^{b(1-\alpha)} K^\alpha L \\
 &= A^{1-\alpha} \left(\frac{K}{N} \right)^{b(1-\alpha)} K^\alpha L
 \end{aligned}$$

$$= a^{1-\alpha} \left(\frac{K}{N} \right)^{b(1-\alpha)} L \left(\frac{K}{L} \right)^{\alpha}$$

If we augment every argument by Z

then we might get

$$= a^{1-\alpha} \left(\frac{ZK}{N} \right)^{b(1-\alpha)} Z L \left(\frac{K}{L} \right)^{\alpha}$$

$$= Z^{b-b\alpha+1} a^{1-\alpha} \left(\frac{K}{N} \right)^{b(1-\alpha)} L \left(\frac{K}{L} \right)^{\alpha}$$

Since $0 < \alpha < 1$

$0 < b\alpha < b$

Then $b - b\alpha > 0$

$b - b\alpha + 1 > 1$

Therefore increasing K and L will increase more than you input it.

